

Mathematical Model & Reactome quirks

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1 Kinetic constants estimation

Notation preliminaries:

- $\mathbb{N}_1 = \mathbb{N} - \{0\}$
- $\mathbb{R}^+ = \{x \mid x \in \mathbb{R} \wedge x \geq 0\}$

1.1 Model definition 4

Let $\mathcal{T} = \{\text{reactant}, \text{product}, \text{enzyme}, \text{activator}, \text{inhibitor}\}$ be the set of roles an entity can undertake in a reaction. In order to simplify notation, the following subsets of $\mathcal{T}$ shall be defined:

- $\mathcal{T}_{\text{stoichiometric}} = \{\text{reactant}, \text{product}\}$ the set of stoichiometric roles
- $\mathcal{T}_{\text{modifier}} = \{\text{enzyme}, \text{activator}, \text{inhibitor}\}$ the set of modifier roles
- $\mathcal{T}_{\text{reagent}} = \{\text{reactant}\} \cup \mathcal{T}_{\text{modifier}}$ the set of reagent (input) roles

Definition 1 (*Biological network*).

A biological network $G = (V, E, c, n)$ is a semi-quantitative graph model where

- $V = X \sqcup R \sqcup C$
 - X is the set of physical entities
 - R is the set of reactions
 - C is the set of compartments in which the entities are located
- $E = E_{\text{stoichiometric}} \sqcup E_{\text{modifier}}$
 - $E_{\text{stoichiometric}} \subseteq X \times R \times \mathcal{T}_{\text{stoichiometric}}$
 - $E_{\text{modifier}} \subseteq X \times R \times \mathcal{T}_{\text{modifier}}$
- $c : X \rightarrow C$ is a function that assigns to each entity a compartment
- $n : E_{\text{stoichiometric}} \rightarrow \mathbb{N}_1$ is the stoichiometry function

Definition 2 (*Scope*).

Given a biological network G let $S = (X_{\text{interest}}, X_{\text{boundary}}, R_{\text{interest}})$ be the scope of a problem in G , where

- $X_{\text{interest}} \subseteq X$ is the set of entities whose reachability must be computed
- $X_{\text{boundary}} \subseteq X$ is the set of entities on the boundary
- $R_{\text{interest}} \subseteq R$ is the set of the only reaction that can be included in the reachability of X_{interest}

Definition 3 (*Scope reachability*).

Given a biological network G and a scope S in G let $G' = (V', E', c', n')$ be the subgraph of G reachable from S .

The set of reactions R' in G' is defined inductively as follows:

- if x is an entity and r is a reaction where $x \in X_{\text{interest}} \wedge r \in R_{\text{interest}} \wedge (x, r, \text{product}) \in E$ then $r \in R'$
- if r_1 is a reaction in R' , r_2 is a reaction in R_{interest} and there is some entity x where $x \notin X_{\text{boundary}}$ and $\{(x, r_1)\} \times \mathcal{T}_{\text{reagent}} \cap E \neq \emptyset$ and $(x, r_2, \text{product}) \in E$ then $r_2 \in R'$

- $X' = \{x \mid \exists r, t \ r \in R' \wedge (x, r, t) \in E_R\}$
- C' is the image of c over X'
- $E' = \{(x, r, t) \mid r \in R' \wedge (x, r, t) \in E\}$
- c' is just like c but restricted to X'
- n' is just like n but restricted to $E'_{\text{stoichiometric}}$

1.2 Reactions rate laws 2

$$\text{regulation}(\mathbf{r}) = \prod_{(\mathbf{x}, \mathbf{r}, \text{activator}) \in \mathbf{E}} \frac{[\mathbf{x}]}{K_A^{\mathbf{x}, \mathbf{r}} + [\mathbf{x}]} \prod_{(\mathbf{x}, \mathbf{r}, \text{inhibitor}) \in \mathbf{E}} \frac{K_I^{\mathbf{x}, \mathbf{r}}}{K_I^{\mathbf{x}, \mathbf{r}} + [\mathbf{x}]} \quad (1)$$

$$\text{rate}(\mathbf{r}) = \text{regulation}(\mathbf{r}) \cdot k_r \prod_{\mathbf{e}=(\mathbf{x}, \mathbf{r}, \text{reactant}) \in \mathbf{E}} [\mathbf{x}]^{n(\mathbf{e})} \quad (2)$$

$$\text{rate}(\mathbf{r}) = [e] \cdot \text{regulation}(\mathbf{r}) \cdot \frac{N}{D} \quad (3)$$

$$N = k_r \prod_{\mathbf{e}=(\mathbf{x}, \mathbf{r}, \text{reactant}) \in \mathbf{E}} \left(\frac{[\mathbf{x}]}{K_m^{\mathbf{x}, \mathbf{r}}} \right)^{n(\mathbf{e})} \quad (4)$$

$$D = \text{to define...} \quad (5)$$

In the “Systems Biology” book (Klipp, Edda, Kowald, Axel, pages 49-50) it says that the the denominator D can be one of the following:

1. Power-law modular rate law: $D = 1$
2. Common modular rate law (Equation 5)
3. Simultaneous binding modular rate law

$$D = \prod_{\substack{(\mathbf{x}, \text{reactant}, \mathbf{n}) \\ \in \\ f_{\mathbf{R}}(\mathbf{r})}} \left(1 + \frac{[\mathbf{x}]}{K_m^{\mathbf{x}, \mathbf{r}}} \right)^{\mathbf{n}} \prod_{\substack{(\mathbf{x}, \text{product}, \mathbf{n}) \\ \in \\ f_{\mathbf{R}}(\mathbf{r})}} \left(1 + \frac{[\mathbf{x}]}{K_m^{\mathbf{x}, \mathbf{r}}} \right)^{\mathbf{n}} \quad (6)$$

4. Direct binding modular rate law:

$$D = 1 + \prod_{\substack{(\mathbf{x}, \text{reactant}, \mathbf{n}) \\ \in \\ f_{\mathbf{R}}(\mathbf{r})}} \left(\frac{[\mathbf{x}]}{K_m^{\mathbf{x}, \mathbf{r}}} \right)^{\mathbf{n}} + \prod_{\substack{(\mathbf{x}, \text{product}, \mathbf{n}) \\ \in \\ f_{\mathbf{R}}(\mathbf{r})}} \left(\frac{[\mathbf{x}]}{K_m^{\mathbf{x}, \mathbf{r}}} \right)^{\mathbf{n}} \quad (7)$$

5. Force-dependent modular rate law:

$$D = \sqrt{\prod_{\substack{(\mathbf{x}, \text{reactant}, \mathbf{n}) \\ \in \\ f_{\mathbf{R}}(\mathbf{r})}} \left(1 + \frac{[\mathbf{x}]}{K_m^{\mathbf{x}, \mathbf{r}}} \right)^{\mathbf{n}} \prod_{\substack{(\mathbf{x}, \text{product}, \mathbf{n}) \\ \in \\ f_{\mathbf{R}}(\mathbf{r})}} \left(1 + \frac{[\mathbf{x}]}{K_m^{\mathbf{x}, \mathbf{r}}} \right)^{\mathbf{n}}} \quad (8)$$

Definition 4 (*Biological model satisfiability problem*).

Given a biological network $\mathbf{G}$ let $\mathbf{T}, \mathbf{I}$ be such that:

- $\mathbf{T}$ is the time horizon for the simulation
- TODO: tollerances for kinetic constants of the same reaction
- TODO: tollerances for kinetic constants of reversible reactions
- $\mathbf{I}$ is the set of indicators to measure
 - given a trajectory
- TODO: additional constrain

Let $\langle \theta, D, C \rangle$ be the satisfiability problem where

$$\begin{aligned} \theta = & \{t_r \mid r \in \mathbf{R}\} \\ & \cup \{t_x \mid x \in \mathbf{X}\} \\ & \cup \{t_A^{x,r} \mid (x, r, \text{activator}) \in \mathbf{E}\} \\ & \cup \{t_I^{x,r} \mid (x, r, \text{inhibitor}) \in \mathbf{E}\} \\ & \cup \{t_{m,+}^{x,r} \mid r \text{ involves an enzyme and } x \text{ is a reactant of } r\} \\ & \cup \{t_{m,-}^{x,r} \mid r \text{ involves an enzyme and } x \text{ is a product of } r\} \\ & \cup \{x_{\text{avg}} \mid x \in \mathbf{X}\} \end{aligned} \quad (9)$$

The respective kinetic constants are defined as 10^t , for example

- $k_A^{x,r} = 10^{t_A^{x,r}}$

$$\begin{aligned} D = & \{D_{t_x} \mid D_{t_x} = [0, 1]\} \\ & \cup \{\text{all other domains are } [-6, 6]\} \end{aligned} \quad (10)$$

Constraints

$$\forall x \in \mathbf{X} \quad 0 \leq [x] \leq 1 \quad (11)$$

$$\forall x \in \mathbf{X}_{\text{avg}} \quad [x](\varphi \cdot T) - [x](T) = 0 \text{ (stability)} \quad (12)$$

$$\forall x_{\text{avg}} \in \mathbf{X}_{\text{avg}} \quad [x_{\text{avg}}] \in [\mathbf{v}(x) - \varepsilon, \mathbf{v}(x) + \varepsilon] \quad (13)$$

$$\begin{aligned} & \forall k_1^r, k_2^r \in \mathbf{K}, x \in \mathbf{X} \\ & \text{if } x \text{ is a modifier of } k_{r_2} \text{ and is produced by } k_1^r \text{ then } k_{r_1} < k_{r_2} \end{aligned} \quad (14)$$

Definition 5 (*Optimization problem*).

Basically take the satisfiability problem above and turn it in a “single objective optimization problem”

Are there reversible reactions where no enzyme is involved?

Is it more common for a reaction to be reversible or not? Are there reversible reactions with different objects?

Is there a reaction which involves both set and non-set physical entities?

How should the compartments of sets be handled? the set should be split into its members, thus each member has its compartments.

Are there reactions with multiple compartments? yes, but they are not a problem for SBML.

What happens when a physical entity is both input and output of a reaction?

How should modifiers which are complexes be handled? this one is hard (both active unit and direct)

Can an entity be both an enzyme and a regulator in a reaction? yes, yes it can.

Do all strange behaviours stem from sets?

Does inferredTo need to be traversed?

Are there inferredTo reactions within the same species?

Is the referenceEntity useful?

Is there a way to model concentrations between in [0,1] directly in libroadrunner?

Is there a reaction which takes an enzyme (catalyst activity) and gives out a complex containing the enzyme?

Is there a reaction with multiple catalyst activities? Yes

Are there reactions without pathways?

•

```
MATCH (reaction:ReactionLikeEvent)
WHERE NOT EXISTS {
    MATCH (pathway:Pathway)--(reaction)
}
RETURN COUNT (DISTINCT reaction) // 2247
```

Are there reactions with multiple pathways?

```
MATCH
    (reaction:ReactionLikeEvent)--(pathway1:Pathway),
    (reaction:ReactionLikeEvent)--(pathway2:Pathway)
WHERE pathway1 <> pathway2
RETURN COUNT(DISTINCT reaction) // 4633
```

```

MATCH
  (reaction:Pathway)<-[:hasEvent]-(pathway1:Pathway),
  (reaction:Pathway)<-[:hasEvent]-(pathway2:Pathway)
CALL
  apoc.path.subgraphNodes(
    pathway,
    {
      relationshipFilter: "hasEvent",
    }
  )
YIELD node

WHERE pathway1 <> pathway2
AND EXISTS {
  MATCH (pathway1)-[:hasEvent *]-(pathway2)
}
RETURN COUNT(DISTINCT reaction)

```

```

MATCH (pathway:Pathway)
CALL
  apoc.path.subgraphNodes(
    pathway,
    {
      relationshipFilter: "hasEvent>",
      labelFilter: ">Pathway"
    }
  )
YIELD node
WITH pathway, COLLECT(DISTINCT node) AS reachablePathways
WHERE NOT pathway IN reachablePathways
RETURN COUNT (DISTINCT pathway) // 8274

```

```

MATCH (pathway:Pathway { :9612973 })
CALL
  apoc.path.subgraphNodes(
    pathway,
    {
      relationshipFilter: "hasEvent>",
      labelFilter: ">Pathway"
    }
  )
YIELD node
WITH pathway, COLLECT(DISTINCT node) AS reachablePathways
RETURN pathway.dbId, reachablePathways // 8274

```

Are there reversible reactions which do not share pathways?

```

MATCH
  (reaction1:ReactionLikeEvent)-->(:CatalystActivity)--
>(:PhysicalEntity)<--(:CatalystActivity)<--
  (reaction2:ReactionLikeEvent)
WHERE
  reaction1 <> reaction2
RETURN COUNT(DISTINCT reaction1)

MATCH (reaction:ReactionLikeEvent)
OPTIONAL MATCH path1 = (reaction)-[:input]->()
OPTIONAL MATCH path2 = (reaction)-[:output]->()
OPTIONAL MATCH path3 = (reaction)-->(:CatalystActivity)--
>(:PhysicalEntity)
OPTIONAL MATCH path4 = (reaction)-->(:Regulation)--
>(:PhysicalEntity)
WHERE reaction.dbId IN [113411, 112392]
RETURN path1, path2, path3, path4

OPTIONAL MATCH (reaction)-[:input]->(entity1)
OPTIONAL MATCH (reaction)-[:output]->(entity2)
OPTIONAL MATCH (reaction)-->(:CatalystActivity)--
>(entity3:PhysicalEntity)
OPTIONAL MATCH (reaction)-->(:PositiveRegulation)--
>(entity4:PhysicalEntity)
OPTIONAL MATCH (reaction)-->(:NegativeRegulation)--
>(entity5:PhysicalEntity)
WITH
  reaction,
  entity1,
  entity2,
  entity3,
  entity4,
  entity5
  ORDER BY reaction, entity1.dbId, entity2.dbId, entity3.dbId,
  entity4.dbId, entity5.dbId
WITH
  reaction,
  COLLECT(DISTINCT entity1) AS e1,
  COLLECT(DISTINCT entity2) AS e2,
  COLLECT(DISTINCT entity3) AS e3,
  COLLECT(DISTINCT entity4) AS e4,
  COLLECT(DISTINCT entity5) AS e5
WITH COLLECT({r: reaction, e1: e1, e2: e2, e3: e3, e4: e4, e5:
e5}) AS rects
UNWIND rects AS r1
UNWIND rects AS r2
WITH r1, r2
WHERE
  r1.r.dbId < r2.r.dbId
  AND r1.e1 = r2.e2
  AND r1.e2 = r2.e1
  AND r1.e3 = r2.e3
  AND r1.e4 = r2.e4
  AND r1.e5 = r2.e5

```

```

MATCH (object)
WHERE
    size(apoc.coll.intersection(["PhysicalEntity"],
labels(object))) <> 0
OPTIONAL MATCH (object)-[relation]->(:PhysicalEntity)
WITH object, type(relation) as type ORDER BY type
WITH DISTINCT labels(object) AS labels, COLLECT(DISTINCT type)
AS types
RETURN labels, types
ORDER BY size(labels), labels, size(types), types

```

```

MATCH (object)
WHERE
    size(apoc.coll.intersection(["PhysicalEntity"],
labels(object))) <> 0
OPTIONAL MATCH (object)-[relation]->(entity:PhysicalEntity)
WITH
    object,
    type(relation) as type,
    apoc.coll.subtract(
        labels(entity),
        ["DatabaseObject", "PhysicalEntity", "Trackable",
"Deletable"]
    ) AS entityLabels
ORDER BY type, entityLabels
WITH DISTINCT labels(object) AS labels, COLLECT(DISTINCT [type,
entityLabels]) AS types
RETURN labels, types
ORDER BY size(labels), labels, size(types), types

```

```

MATCH (object)
WHERE
    size(apoc.coll.intersection(["PhysicalEntity"],
labels(object))) <> 0
UNWIND keys(object) AS key
WITH object, key ORDER BY key
WITH DISTINCT labels(object) AS labels, COLLECT(DISTINCT key)
AS keys
RETURN labels, keys
ORDER BY size(labels), labels, size(keys), keys

```

```

OPTIONAL MATCH (reaction)-[relation]-(entity)
WHERE
    NOT type(relation) IN [
        "precedingEvent", "negativePrecedingEvent",
        "normalReaction",
        "inferredTo", "reverseReaction"
    ]
    AND size(apoc.coll.intersection(
        [
            "ControlledVocabulary", "ReviewStatus",
            "UpdateTracker",
            "InstanceEdit", "Publication",
            "LiteratureReference",
            "ExternalOntology", "EvidenceType", "URL",
            "Pathway",
            "Figure", "Anatomy", "Book", "MetaDatabaseObject",
            "DatabaseIdentifier", "RegulationReference",
            "CatalystActivityReference",
            "EntityFunctionalStatus",
            "DrugActionType", "GO_CellularComponent",
            "GO_Term",
            "GO_BiologicalProcess", "Species", "Taxon",
            "Summation",
            "Compartment"
        ],
        labels(entity)
    )) = 0
WITH
    reaction,
    apoc.coll.subtract(
        labels(entity),
        [
            "DatabaseObject", "Deletable", "Trackable",
            "ReactionLikeEvent", "Event",
            "EntitySet", "CandidateSet", "DefinedSet",
            "SimpleEntity",
            "EntityWithAccessionedSequence",
            "GenomeEncodedEntity",
            "OtherEntity", "Complex", "Disease",
            "Polymer", "ProteinDrug", "Cell", "Requirement",
            "Drug", "RNADrug", "ChemicalDrug",
            "NegativeRegulation", "PositiveRegulation",
            "NegativeGeneExpressionRegulation",
            "PositiveGeneExpressionRegulation"
        ]
    ) as labels
UNWIND labels AS label
WITH reaction, label ORDER BY label
WITH DISTINCT reaction, COLLECT(DISTINCT label) AS types
RETURN
    apoc.coll.subtract(
        labels(reaction),
        ["DatabaseObject", "Deletable", "Trackable"]
    ) AS labels,
    types,
    COUNT(*) AS cnt

```

2 Reactome quirks

2.1 Compartments

2.1.1 Physical entities with multiple compartments

Because the functions of biologic molecules critically depend on their subcellular locations, chemically identical entities located in different compartments are represented as distinct physical entities. The problem is that many physical entities are related to multiple compartments, even though they are already an instance of an entity related to a compartment.

```
MATCH
  (physicalEntity:PhysicalEntity),
  (physicalEntity)-[:compartment]->(compartment1:Compartment),
  (physicalEntity)-[:compartment]->(compartment2:Compartment)
WHERE compartment1.dbId < compartment2.dbId
OPTIONAL MATCH (compartment1)-[relation]-(compartment2)
WITH
  DISTINCT physicalEntity,
  COLLECT(
    DISTINCT CASE
      WHEN relation IS NOT NULL THEN TYPE(relation)
      ELSE "none"
    END
  ) AS links
RETURN
  apoc.coll.subtract(
    LABELS(physicalEntity),
    [
      "DatabaseObject", "Deletable", "Trackable",
      "PhysicalEntity", "EntitySet"
    ]
  ),
  links,
  COUNT(*)
```

Listing 1: Type of links between compartments of physical entities with multiple compartments.

entity type	compartments link types	#
<u>CandidateSet</u>	none	286
<u>CandidateSet</u>	<i>surroundedBy</i>	29
<u>CandidateSet</u>	none, <i>surroundedBy</i>	3
<u>DefinedSet</u>	none	201
<u>DefinedSet</u>	<i>instanceOf</i>	1
<u>DefinedSet</u>	<i>surroundedBy</i>	129
<u>Polymer</u>	<i>surroundedBy</i>	7

If a Polymer involves multiple compartments (or, more generally, a PhysicalEntity which is not an EntitySet) then:

1. Add new physical entities, one for each compartment
2. Add reactions with infinite speed that move substance between compartments

```

MATCH (entitySet:EntitySet)
WHERE EXISTS {
    MATCH
        (entitySet)-[:hasMember|hasCandidate]->(physicalEntity:PhysicalEntity)
    WHERE
        NOT EXISTS {
            MATCH
                (entitySet)-[rel1:compartment|includedLocation]-
                (compartment:Compartment)
            -[rel2:compartment|includedLocation]-(physicalEntity)
        }
}
RETURN COUNT(DISTINCT entitySet) // 0

```

Listing 2: There are no instances of EntitySet which don't share compartments with their members.

If the relation type `includedLocation` is not used in Listing 2 then there is an instance of a Complex that doesn't share its compartment with the set it's placed in.

2.1.2 Physical entities without compartments

Some physical entities do not have compartments. The good news is that these entities are only 25 and are of type Cell (*the total number of cells in reactome is 25*). Moreover, all the reactions that involve cells are of type CellDevelopmentStep

```

MATCH (physicalEntity:PhysicalEntity)
WHERE NOT EXISTS {
    MATCH (physicalEntity)-[:compartment]->(:Compartment)
}
OPTIONAL MATCH path = (physicalEntity)--(:ReactionLikeEvent)
RETURN physicalEntity, path

```

Listing 3: Physical entities without any *compartment* and their reactions.

To handle this case it's enough to add a default compartment.

2.1.3 On the compartments of reactions

```

MATCH (reaction:ReactionLikeEvent)
WHERE EXISTS {
    MATCH
        (reaction)--(compartment1:Compartment),
        (reaction)--(compartment2:Compartment)
    WHERE
        compartment1 <> compartment2
}
RETURN COUNT (DISTINCT reaction) // 36322 (out of 93672)

```

Listing 4: Reactions connected to multiple compartments

The fact that reactions are connected to multiple compartments is not a problem! In SBML the compartment is optional for reactions, as the simulators should be able to deduce the compartment of the reaction based on the compartment of its substrate.

2.2 Reactions

2.2.1 Reactions with multiple enzymes

```
MATCH (reaction:ReactionLikeEvent)
WHERE EXISTS {
    MATCH
        (reaction)--(activity1:CatalystActivity),
        (reaction)--(activity2:CatalystActivity)
    WHERE
        activity1 <> activity2
}
RETURN COUNT(DISTINCT reaction) // 17 reactions out of 93672
```

Listing 5: Reactions with multiple CatalystActivity

The interesting fact about these reactions is that all of them are of type BlackBoxEvent. It might be interesting to add an option to exclude certain types of reactions. In this case a new reaction for each enzyme can be created.

2.2.2 On the reversibility of reactions

```
MATCH (:ReactionLikeEvent)-[relation]-(:ReactionLikeEvent)
RETURN DISTINCT type(relation)
```

Listing 6: Types of links between reactions:
[precedingEvent, inferredTo, normalReaction, reverseReaction]

Explanation:

- *precedingEvent*: useful for visualization
- *inferredTo*: reactions inferred from reactions in a different species
- *normalReaction*: it reference to the normal version of a FailedReaction
- *reverseReaction*: reference to another reaction which is the reverse

```
MATCH (reaction:ReactionLikeEvent)-[:reverseReaction]->(:ReactionLikeEvent)
RETURN COUNT(DISTINCT reaction) // 112
```

Listing 7: Number of reactions which have a reference to a reverse reaction

Such reactions are marked as (Reversible) in the PathwayBrowser

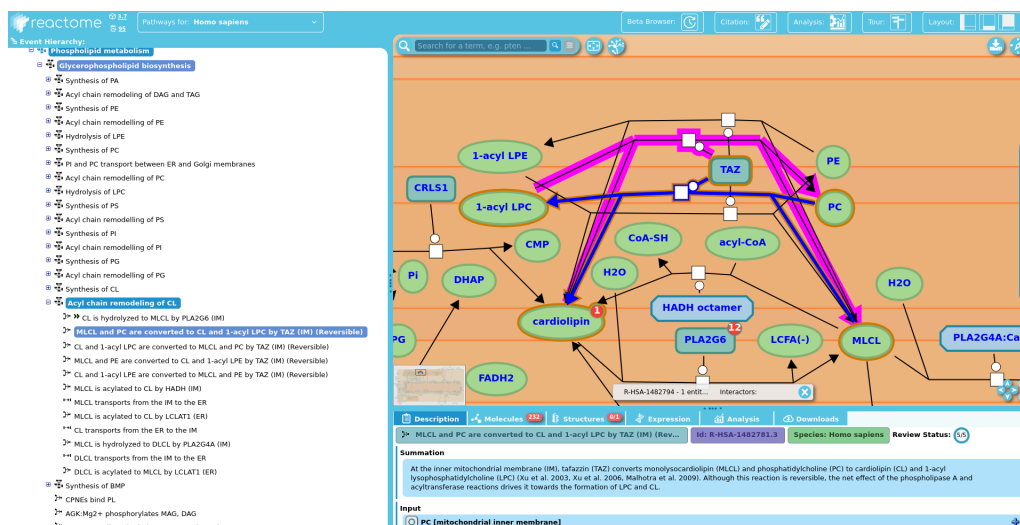


Figure 1: Reversible reaction in the PathwayBrowser: [link to view](#), [link to reaction page](#)

```
MATCH (reaction:ReactionLikeEvent)
UNWIND keys(reaction) AS key
RETURN DISTINCT key
```

Listing 8: Attributes of a generic ReactionLikeEvent.

The attributes are: schemaClass, speciesName, isInDisease, releaseDate, displayName, stIdVersion, dbId, name, isChimeric, category, stId, isInferred, oldStId, releaseStatus, definition, systematicName.

Non of them give information about the “reversibility” of the reaction.

In the official exporter of Reactome the reversibility option is set to `false` by default: [link to code](#).

I couldn't find no other way to determine if a reaction is reversible or not, so only those 112 reaction can be considered reversible.

2.3 Roles physical entities have in reactions

2.3.1 Reactions classification based on the presence of reactants, products, enzymes and regulators

```
MATCH
  (reaction:ReactionLikeEvent)
OPTIONAL MATCH (reaction)-[relation]-(entity)
WHERE
  NOT type(relation) IN [
    "precedingEvent", "negativePrecedingEvent", "normalReaction",
    "inferredTo", "reverseReaction"
  ]
  AND size(apoc.coll.intersection(
    [
      "ControlledVocabulary", "ReviewStatus", "UpdateTracker",
      "InstanceEdit", "Publication", "LiteratureReference",
      "ExternalOntology", "EvidenceType", "URL", "Pathway",
      "Figure", "Anatomy", "Book", "MetaDatabaseObject",
      "DatabaseIdentifier", "RegulationReference",
      "CatalystActivityReference", "EntityFunctionalStatus",
      "DrugActionType", "GO_CellularComponent", "GO_Term",
      "GO_BiologicalProcess", "Species", "Taxon", "Summation",
      "Compartment"
    ],
    labels(entity)
  )) = 0
WITH
  reaction,
  apoc.coll.subtract(
    labels(entity),
    [
      "DatabaseObject", "Deletable", "Trackable",

      "ReactionLikeEvent", "Event",

      "EntitySet", "CandidateSet", "DefinedSet", "SimpleEntity",
      "EntityWithAccessionedSequence", "GenomeEncodedEntity",
      "OtherEntity", "Complex", "Disease",
      "Polymer", "ProteinDrug", "Cell", "Requirement",
      "Drug", "RNADrug", "ChemicalDrug",

      "NegativeRegulation", "PositiveRegulation",
      "NegativeGeneExpressionRegulation",
      "PositiveGeneExpressionRegulation"
    ]
  ) as labels
UNWIND labels AS label
WITH reaction, label ORDER BY label
WITH DISTINCT reaction, COLLECT(DISTINCT label) AS types
RETURN
  apoc.coll.subtract(
    labels(reaction),
    [
      "DatabaseObject", "Deletable", "Trackable",
      "Event", "ReactionLikeEvent"
    ]
  ) AS labels,
  types,
  COUNT(*) AS cnt
ORDER BY labels DESC, size(types)
```

In the table below reaction types are classified by the roles of the reagents in the reaction (where PhysicalEntity groups inputs and outputs). The rows marked in red indicate the reactions that have regulators (inhibitors and activators) but no enzyme.

<u>Reaction</u>	<u>PhysicalEntity</u>	33477
<u>Reaction</u>	<u>CatalystActivity</u> , <u>PhysicalEntity</u>	43889
<u>Reaction</u>	<u>PhysicalEntity</u> , <u>Regulation</u>	2064
<u>Reaction</u>	<u>CatalystActivity</u> , <u>PhysicalEntity</u> , <u>Regulation</u>	3244
<u>Polymerisation</u>	<u>PhysicalEntity</u>	178
<u>Polymerisation</u>	<u>CatalystActivity</u> , <u>PhysicalEntity</u>	36
<u>Polymerisation</u>	<u>PhysicalEntity</u> , <u>Regulation</u>	15
<u>Polymerisation</u>	<u>CatalystActivity</u> , <u>PhysicalEntity</u> , <u>Regulation</u>	10
<u>FailedReaction</u>	<u>PhysicalEntity</u>	171
<u>FailedReaction</u>	<u>CatalystActivity</u> , <u>PhysicalEntity</u>	271
<u>FailedReaction</u>	<u>PhysicalEntity</u> , <u>Regulation</u>	5
<u>FailedReaction</u>	<u>CatalystActivity</u> , <u>PhysicalEntity</u> , <u>Regulation</u>	2
<u>Depolymerisation</u>	<u>PhysicalEntity</u>	3
<u>Depolymerisation</u>	<u>CatalystActivity</u> , <u>PhysicalEntity</u>	25
<u>CellDevelopmentStep</u>	<u>PhysicalEntity</u>	3
<u>CellDevelopmentStep</u>	<u>PhysicalEntity</u> , <u>Regulation</u>	19
<u>BlackBoxEvent</u>	<u>PhysicalEntity</u>	5135
<u>BlackBoxEvent</u>	<u>CatalystActivity</u> , <u>PhysicalEntity</u>	2746
<u>BlackBoxEvent</u>	<u>PhysicalEntity</u> , <u>Regulation</u>	2144
<u>BlackBoxEvent</u>	<u>CatalystActivity</u> , <u>PhysicalEntity</u> , <u>Regulation</u>	235